PHASE 8 OF 8

Security, Roadmap & Conclusion

What was fixed, what remains, how to harden the app, and where EduGenie can grow next.

Fixes Applied | Known Gaps | Security | Roadmap

8.1 Fixes Applied to the Original Draft

According to the project's README, these issues were found and corrected in this version:

File	What was fixed
`quiz_module.py`	Added missing imports (`re`, `json`); fixed the Markdown-fence-stripping regex; added JSON shape validation.
`qna.py` / `summary_module.py` / `learning_path.py`	Added the Gemini client import and API-key setup via .env; added a clear error message when the key is missing.
`explanation_module.py`	Model and tokenizer now load once and are cached; generation runs inside `torch.no_grad()`.
`main.py`	Standardized endpoint paths, added input validation on GET and POST routes, added the / route serving the frontend.
`templates/index.html`	Completed the unfinished HTML/JS so every form calls its endpoint and shows results, including quiz auto-grading.

8.2 Known Gaps & Things to Double-Check

Gap	Risk	Recommended action
`templates/index.html` and `static/style.css` were not part of the files reviewed	App will not start or render	Confirm both exist before running
`gemini-3.6-flash` used across four modules	Errors if unavailable to your key	Run `check_models.py`; centralize the name
No .env.example file, although the README references one	New users unsure which variable to set	Add the template shown in Phase 3
No request timeout on Gemini calls	Endpoints may hang indefinitely	Set a timeout on every Gemini call

8.3 Security & Configuration Considerations

Area	Recommendation	Priority
Secrets	Never commit .env; add it to .gitignore. Rotate the Gemini key periodically and use a secrets manager in production.	**High**
Timeouts	Add request timeouts so one slow call cannot tie up a worker indefinitely.	**High**
Public exposure	Before deploying beyond localhost, restrict CORS and add authentication — no route requires a login today.	**High**
Input size	Cap the length of pasted text for Summarize and Quiz to avoid excessively large prompts.	**Medium**
Validation	Current checks confirm fields are non-empty; extend to type and length checks.	**Medium**

8.4 Pre-Deployment Checklist

	Item
☐	API key stored outside the codebase; .env ignored by git
☐	Timeouts added to all Gemini calls
☐	CORS restricted to known origins; authentication in place
☐	Input length limits enforced on text fields
☐	Automated tests passing for quiz parsing and route validation
☐	Model weights pre-downloaded on the server to avoid slow first requests

8.5 Suggested Future Enhancements

Horizon	Enhancement
Now	Request timeouts; a .env.example file; automated tests around quiz JSON parsing
Next	Loading indicator in the UI; caching repeated Explain requests for the same topic
Later	Persist quiz results and learning-path history per user for progress tracking

8.6 Conclusion

EduGenie demonstrates a practical pattern for AI-assisted study tools: keep the frontend simple, let a FastAPI backend own all external calls, and split work between a cloud model (Gemini) for open-ended reasoning and a local model for a single, latency-sensitive feature. The one-file-per-feature module boundaries make it straightforward to extend: adding a capability generally means adding one new module and one new route.

Key takeaways

- One module per feature keeps the codebase easy to navigate and extend.
- The cloud/local split is a deliberate latency and cost trade-off, not an inconsistency.
- Input validation and JSON-shape checking keep a model-backed API from crashing on a bad response.
- The biggest near-term risks are the missing request timeouts and the absence of automated tests.

Appendix A — Glossary

Term	Meaning
FastAPI	Python web framework used to define and serve the API routes in main.py
Gemini API	Google's cloud-hosted large language model, used for Q&A, summaries, quizzes and learning paths
LaMini-Flan-T5-783M	Smaller open-source language model run locally for the Explain feature
Uvicorn	ASGI server that runs the FastAPI application
Jinja2	Template engine used to render index.html
.env file	Local file holding secrets such as GEMINI_API_KEY, kept out of version control

Appendix B — Quick Reference Commands

```
python -m venv venv                # create virtual environment
source venv/bin/activate           # activate (Windows: venv\Scripts\activate)
pip install -r requirements.txt    # install dependencies
uvicorn main:app --reload          # run the app
python check_models.py             # list Gemini models for your key
```

Appendix C — Sample requirements.txt

```
fastapi
uvicorn[standard]
jinja2
python-multipart
google-genai
transformers
torch
python-dotenv
```